

DANIEL ROCCA

Lead AI & Full Stack Software Engineer

+1 (213) 297-0175 | contact@danielrocca.dev | <https://danielrocca.dev> | Blackwood, NJ

SUMMARY

Senior Full-Stack & AI Engineer with 15+ years of software engineering experience building production-grade applications, cloud platforms, and intelligent systems. Deep expertise in Python-based AI development, Agentic AI, LLM orchestration, RAG, FastAPI, APIs, distributed systems, and cloud-native architecture, with hands-on experience delivering enterprise AI workflows using LangGraph, OpenAI, Gemini, and Claude. Experienced in taking AI solutions from architecture and prototyping through evaluation, deployment, observability, and production support. Combines strong Python/backend engineering with full-stack, data, and cloud experience to build scalable AI applications, multi-agent workflows, retrieval systems, and automation platforms designed for reliability, performance, and measurable business impact.

EXPERIENCE

Lead AI & Full-Stack Software Engineer

Origami Studios

11/2023 - 12/2025 | Parsippany, NJ

- Led architecture and full-stack delivery across four product streams for MyTutor's UK and U.S. platforms, building scalable backend services, APIs, automation workflows, and cloud-native application capabilities.
- Helped evolve the platform toward AWS microservices, serverless components, and event-driven architecture, creating a stronger foundation for data-intensive workflows, integrations, and future AI capabilities.
- Delivered production systems spanning backend services, scheduling automation, authentication, analytics, payments, virtual classrooms, and third-party APIs, supporting complex multi-role workflows.
- Contributed to a platform serving 250K+ learners, 1,500+ schools, and 4M+ lessons, working across architecture, implementation, testing, deployment, and production reliability.

Full-Stack AI Software Engineer

Walmart Global Tech

01/2020 - 09/2023 | Hoboken, NJ

- Led production delivery of an Agentic AI financial reconciliation platform, automating transaction matching, exception handling, human review, and continuous-learning workflows across high-volume financial operations.
- Architected a Python/LangGraph Supervisor-Worker multi-agent framework with confidence-based self-correction and semantic tool routing, improving tool-selection accuracy from 70% to 95% while reducing token consumption by approximately 80%.
- Designed an intelligent reconciliation pipeline combining structured processing, clustering, deterministic rules, and LLM agents, achieving 90%+ matching accuracy across 10,000+ daily transactions.
- Built production AI capabilities spanning human-in-the-loop learning, API/service integration, evaluation, guardrails, observability, and failure recovery, owning delivery from architecture through rollout and operational support.

Software Development Engineer II

Amazon

09/2017 - 01/2020 | Remote

- Modernized business-critical Amazon Payments workloads by migrating Oracle-based systems to DynamoDB, strengthening the data foundation for scalable, cloud-native processing.
- Redesigned backend data-access patterns and distributed service boundaries for high-volume NoSQL workloads, improving scalability, performance, and maintainability.
- Refactored production services around modular, testable, and fault-isolated architectures, experience directly applicable to reliable Python AI services and orchestration layers.
- Strengthened CI/CD, automated testing, deployment reliability, and production operations, supporting safe delivery of distributed systems at enterprise scale.

Full-Stack & Integration Engineer

Cleo

07/2015 - 08/2017 | Blackwood, NJ

- Designed and delivered enterprise B2B integration services connecting customer platforms with suppliers, trading partners, and major retailers through APIs and structured data workflows.
- Built reusable EDI, REST, and SOAP services for document ingestion, transformation, validation, and system-to-system automation across complex enterprise environments.
- Developed full-stack integration tooling for transaction processing, operational data, external APIs, and workflow management, strengthening experience with data-driven backend systems.
- Supported the full production lifecycle from implementation and testing through deployment, troubleshooting, and operational support.

Full-Stack Software Engineer

FreedomPay

03/2012 - 12/2014 | Philadelphia, PA

- Developed backend and full-stack capabilities for an enterprise payments platform, including transaction workflows, APIs, data integrations, and merchant-facing applications.
- Built reusable API and integration services connecting payment systems, POS platforms, merchant applications, and external services.
- Supported high-volume transaction-processing systems with strong requirements around data integrity, availability, performance, and production reliability.
- Worked across implementation, integration, testing, deployment, and operational support, establishing a strong foundation in production backend engineering and distributed application delivery.

Software Developer & Scrum Master

Elsevier

10/2011 - 01/2012 | Philadelphia, PA

- Worked directly on application development, debugging, testing, and technical problem solving while coordinating delivery across engineering and product stakeholders.
- Translated business requirements into structured technical tasks, user stories, and implementation priorities, keeping development aligned with measurable business needs.
- Coordinated sprint planning, backlog refinement, testing, and release activities across the software development lifecycle.
- Helped strengthen engineering quality, release readiness, collaboration, and delivery predictability through disciplined Agile development practices.

EDUCATION

Master's Degree, Computer Science - Princeton University | 2007 - 2009

Bachelor's Degree, Computer Science - Stevens Institute of Technology | 2003 – 2007

TECH STACK

Languages: Python • TypeScript • JavaScript • Java • SQL • PHP

AI & GenAI: Agentic AI • LLMs • RAG • LangGraph • Multi-Agent Systems • OpenAI • Gemini • Claude • Prompt Engineering • Human-in-the-Loop AI • Structured Outputs

Python & Backend: FastAPI • REST APIs • Microservices • Async Processing • API Integration • Backend Services

Data & Retrieval: PostgreSQL • DynamoDB • Oracle • SQL/NoSQL • Data Processing • Semantic Retrieval • Embeddings • Vector Search

Cloud & DevOps: AWS • Kubernetes • Docker • CI/CD • Serverless • Event-Driven Architecture • Observability • Automated Testing

Frontend: React • TypeScript • JavaScript

Architecture: Distributed Systems • Cloud-Native Architecture • AI Orchestration • Production AI • System Design • Evaluation • Guardrails • Reliability Engineering